



Introduction to Embodied AI

Lecture 1: Introduction

Junwei Liang
AI Thrust
HKUST (Guangzhou)

Your Instructor and TAs This Semester (AIAA 4220)



Junwei Liang
junweiliang@hkust-gz.edu.cn
Course lecturer

Our lectures are scheduled on Monday every week from 1:30 pm. to 4:20 pm.

First half: 1:30 to 2:45 pm.

Quiz: 2:30pm.~

Second half: 3:00 to 4:20 pm.

Your TA:

Jiayi Liu
jliu334@connect.hkust-gz.edu.cn

Teng Wang
twang560@connect.hkust-gz.edu.cn

There is a parallel class taught by Prof. Changhao Chen on Wednesday. We will share exactly **the same quizzes (not exactly) and group projects.**

Consider switching to Prof. Chen's class to even the numbers.

Instructor	Quota	Enrol
LIANG, Junwei	50	29
CHEN, Changhao	50	10

Logistics: Part 1

You can download the slides now for today's lecture if you want to take note on it

- Course website
 - <https://hkust-aiaa4220.github.io/fall2025-L01/index.html>
 - You can see the **course schedule, homework links, download slides, policy**, etc.
 - The slides will be available before each class

Introduction to Embodied AI

AIAA 4220, Fall 2025

Instructors: [Junwei Liang](#)



Home Schedule	
Lecture:	Date and Time: 01:30PM - 04:20PM, Monday Location: Rm 149, E1 Websites: Canvas Zoom: 628-334-1826 PW: 12345 (In-person expected) (Recordings will be available after each class)
Course Description:	This course aims to provide students with key principles and algorithms to build modern Embodied AI systems. Key topics include machine perception, embodied navigation and manipulation algorithms. Through this course, students will learn and practice the foundational principles, techniques, and tools to build new Embodied AI systems.
Target Audience/ Prerequisites:	This is a course primarily for junior and senior students.
Course Work:	Grading In-Class Quizzes Project 1: Presentation Grading will be based on 10 in-class quizzes and attendance (30%), one presentation project (20%), one machine perception homework (20%) and one navigation homework (30%). • The in-class quizzes are designed to complement the lectures and making sure you understand the key knowledge about the course. • There are 10 in-class quizzes over the semester. The attendance is taken when you complete the quiz in-class. • All quizzes will be 9 multiple-choice questions. You will have 10 minutes (tentative) to finish. • You will select 1-2 papers from our list of Embodied AI research to present. This should show that you understand the paper, and you could present it to students who do not know the paper well. • Can be done in groups of 1-3 people. • Project 1 is worth 20% credit. You will be graded based on Understanding & Content (40%), Presentation & Communication (30%), Critical Analysis & Discussion (30%), by the teacher, TAs, and peers.

We will go over the course policy later today

Agenda

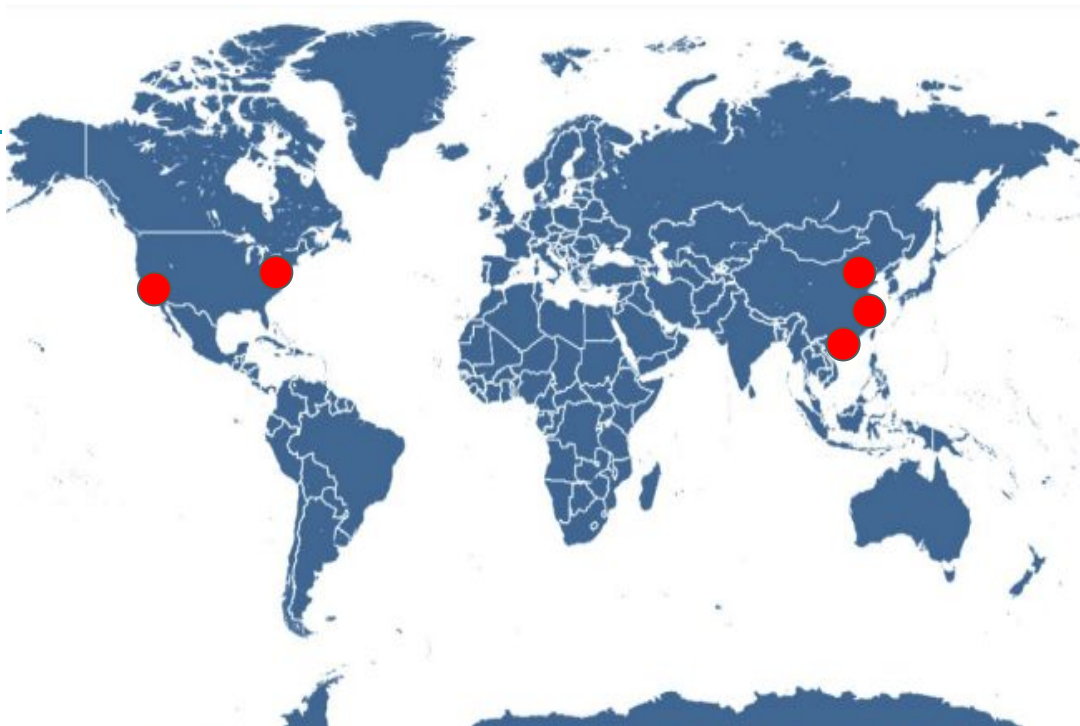
- A bit about me...
- What is Embodied AI?
- History of Robotics
- Why study Embodied AI now?
- Course content overview
- Logistics

A bit about me ...



Junwei Liang
junweiliang@hkust-gz.edu.cn
AIAA 4220 course lecturer

- I'm from Guangzhou
- I studied in Beijing (@RUC), Pittsburgh (@CMU) and interned in Mountain View (@Google)
- I moved to Shanghai (@Tencent)
- And now I'm back!



Our Lab is the leading Embodied AI lab in the greater bay area

- Check our lab website for more cool demos: <https://precognition.team/> Or check out our 小红书!

Navigation & Locomotion

Machine Perception

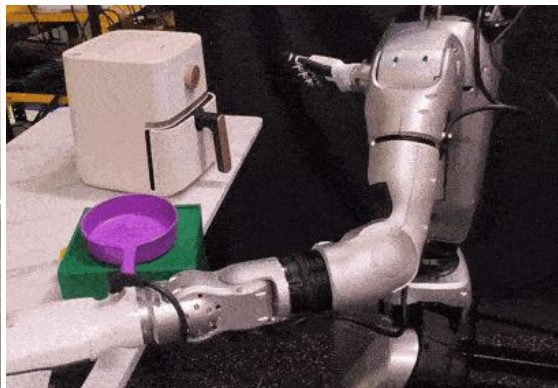
**Embodied
AI**

Manipulation

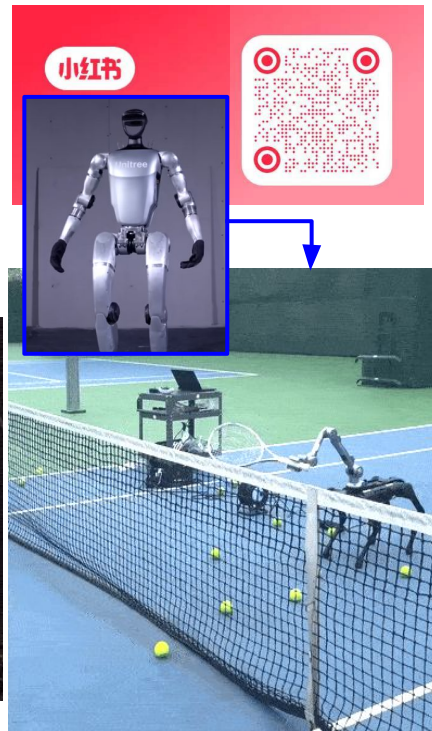
Forecasting



Agile Locomotion



Manipulation with dex hand



Athletic Intelligence

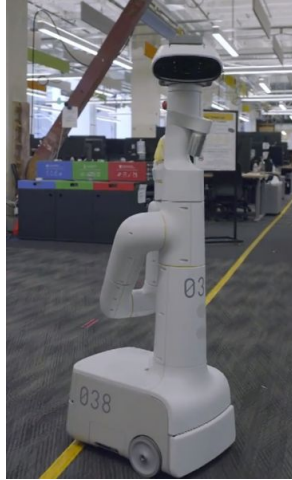
Agenda

- A bit about me...
- **What is Embodied AI?**
- History of Robotics
- Why study Embodied AI now?
- Course content overview
- Logistics

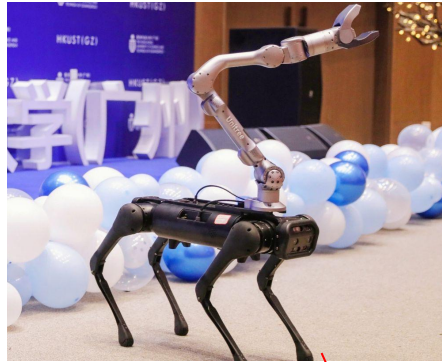
What is Embodied AI?



Boston Dynamics



Everyday Robot



Unitree Robotics



Mobile manipulation demo @MS 4th floor
(Jacobi.ai)

Robots? Two-legged, wheeled, four-legged...

See how they move

* <https://www.youtube.com/watch?v=Gm6dZ1q06ks>

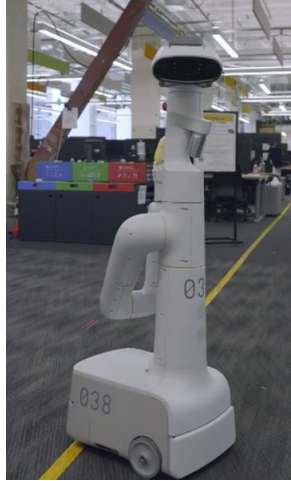
* https://www.youtube.com/watch?v=-e1_QhJ1EhQ

BTW, if you want to do
research with this, let me know

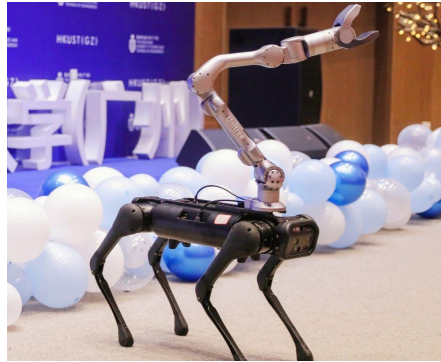
What is Embodied AI?



Boston Dynamics



Everyday Robot



Unitree Robotics



Mobile manipulation demo @MS 4th floor
(Jacobi.ai)

Robots? Any machine that can perform a task automatically!

- Could be based on prespecified rules or through machine learning..

What is Embodied AI?

- Embodied AI is a robot capable of **sensing its environment** and **performing its tasks accordingly** in the real world.
- Key intelligent properties
 - Adaptability (account for noise in the world)
 - Learning from data



Are washing machines considered Embodied AI? No.
There is no environment sensing

Are cat litter machines Embodied AI? Maybe. It detects cats.



What is Embodied AI?

- Embodied AI is the intersection of **machine learning and robotics**
 - In Chinese: “具身智能”
- The study of **training optimal solutions using sensory data** for robots to act and achieve goals
- Perception and decision making are the key
- Possible future application
 - General-purpose home robots!



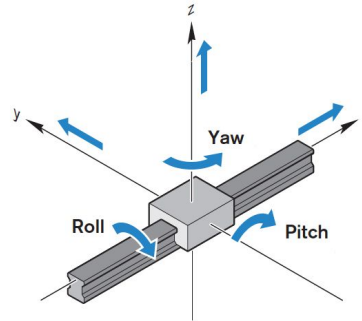
Embodied AI - What for?

- The AI should take over the three “D”s (according to the robotics researcher)
 - The dull
 - Repetitive tasks
 - The dirty
 - The dangerous
 - Warehouse robots
 - Cleaning bots
 - Underwater robots
 - etc.

When Machine Learning is **NOT** Needed

- Industrial Robot

- Populated from 1970s
- Usually a single arm with no mobility
- Move an object with up to 6 DoF
 - Xyz - position
 - pitch, yaw, roll - direction
- Programmed to do repetitive actions (The Dull)
 - Sometime dangerous, lifting heavy stuff
- **Repetitive, non-changing processes**



Define points P1–P5:

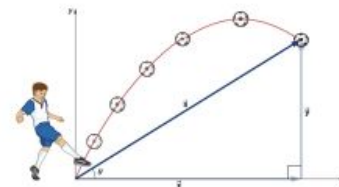
1. Safely above workpiece (defined as P1)
2. 10 cm Above bin A (defined as P2)
3. At position to take part from bin A (defined as P3)
4. 10 cm Above bin B (defined as P4)
5. At position to take part from bin B. (defined as P5)

Define program:

1. Move to P1
2. Move to P2
3. Move to P3
4. Close gripper
5. Move to P2
6. Move to P4
7. Move to P5
8. Open gripper
9. Move to P4
10. Move to P1 and finish

When Machine Learning is **NOT** Needed (for most of the time...)

- Physical processes
 - Kinematics and inverse kinematics of different actuators
 - Meaning: how robots (parts) move to a desired position
 - We don't have to "reinvent the wheel". We have good models of physical processes
 - Boston Dynamics
 - Unitree
 - We can just call their APIs. Although you can still use machine learning to improve
 - With reinforcement learning!
 - Key problems include Sim-to-Real, accounting for the imperfect actuation



When Machine Learning **IS** Needed

- Robots in **unstructured environments**



Structured/known environment



Environment
uncertainty



Object variation

Embodied AI vs. Automation

Special-Purpose Robot Automation

Without machine learning, it is usually called automation, not autonomy



custom-built robots



human expert programming & manual calibration



special-purpose behaviors

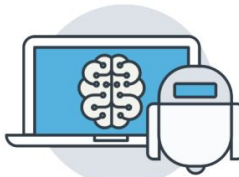
General-Purpose Robot Autonomy



general-purpose robots



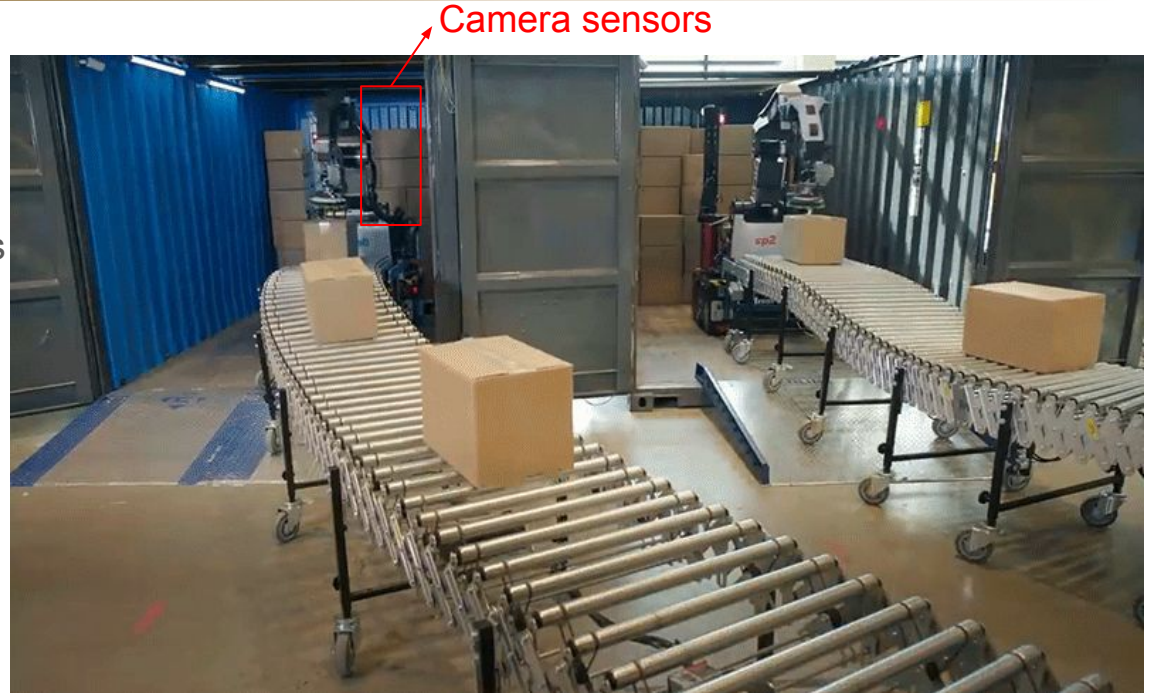
Robot Learning



general-purpose behaviors

Advancement in Unstructured Environment

- (New) Warehouse Robot
 - Boston Dynamics 2022 new product
 - Unloading boxes from trucks to warehouses
 - Adapt to many kinds of boxes
 - Suction gripper



* Learn more about Stretch: <https://www.youtube.com/watch?v=qgGH6Vk8nY8>

Advancement in Unstructured Environment



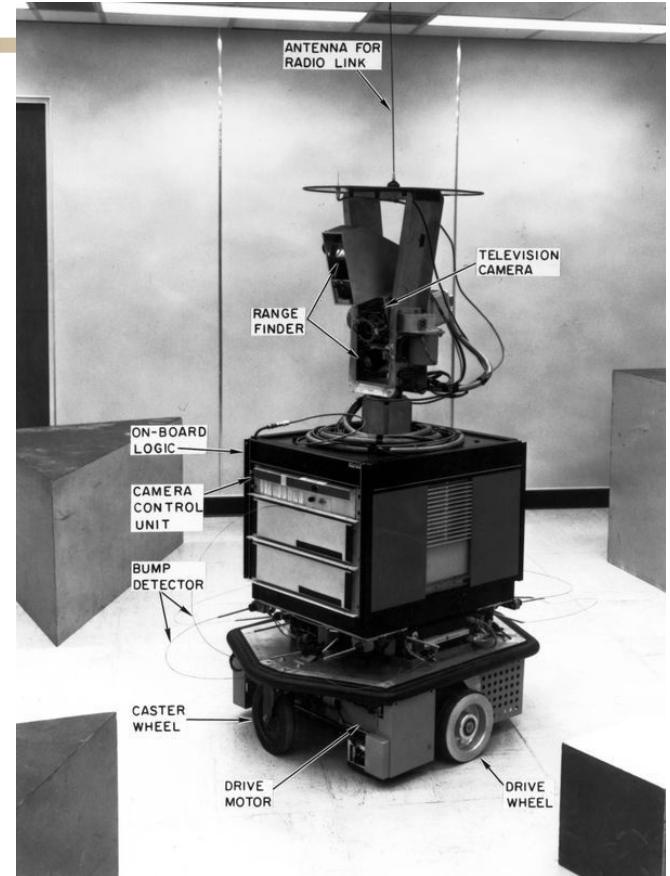
- We are already using some form of Embodied AI products
 - Cleaning robots - key tech: SLAM
 - Pets litter box - pet recognition - infrared and weight sensor
 - Home Security camera - person detection and tracking
 - Self-driving taxi
 - Delivery robots

Agenda

- A bit about me...
- What is Embodied AI?
- **History of Robotics**
- Why study Embodied AI now?
- Course content overview
- Logistics

History of Robotics

- The first “real” robot
 - 1960s - Shakey the robot - A* algorithm
 - Charles Rosen, Stanford Research Institute



History of Robotics

- 1980s - Robotic Arms are populated in Automation



History of Robotics

- 2000s
 - Cleaning robots - Roomba/iRobot - 2002
 - Simple algorithms
 - Simple bump sensor
 - No mapping of the room
 - Random walk angle changing after bumping into an object or wall



Long exposure photo showing path taken by a Roomba as it cleans



History of Robotics

- 2000s
 - 2004 - 2007 - DARPA Grand Challenge
 - First autonomous driving competition
 - CMU won in 2007, finishing a **96 km** urban course in over **4 hours** (avg. **22.53 km/h**)
 - So slow it was like a bicycle ride
 - Already uses modern senses like cameras, lidars and radars*

* <http://www.tartanracing.org/tech.html>

* paper: https://www.ri.cmu.edu/pub_files/2007/4/Tartan_Racing.pdf



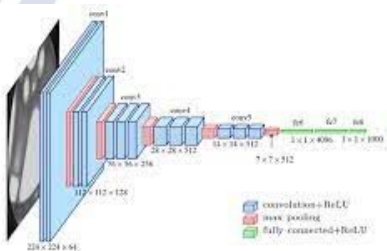
Today's Robotics

- 2010s -
 - Consumer-grade drones - DJI
 - Self-driving cars - Waymo, etc.
 - Highly-agile robots - Boston Dynamics
 - Humanoid robot - Tesla 2022
 - Might be promising, already at GEN 2



Agenda

- A bit about me...
- What is Embodied AI?
- History of Robotics
- **Why study Embodied AI now?**
- Course content overview
- Logistics



Artificial Intelligence

Recent breakthroughs in machine learning, language and computer vision: CLIP, ChatGPT. Expect a good Large Vision-Language Model in recent years



Computing Power

Your smartphone is millions of times more powerful than all of NASA's combined computing in 1969.



Robot Hardware

More reliable and affordable robot hardware. Boston Dynamics' spot costs \$75,000
Unitree has \$10,000 ones



3D Simulators

More open-source,
easy-to-use 3D
simulators.
*CARLA
*NVIDIA Isaac Sim

Now is a good time to study and work on Embodied AI

- Covid-19 makes us realize the importance of Embodied AI that limits human interactions.



By Brianna Wessling | February 6, 2023

ABB uses robots to automate COVID antibody testing

The system, created by ABB and UTMB, can increase the number of neutralizing antibody tests performed from 15 a day to over 1,000 daily.

By Steve Crowe | January 8, 2021

Ava Robotics launches UV disinfection robot

Ava Robotics, the iRobot spinout known for its telepresence robots, is expanding its product portfolio with the introduction of an ultraviolet (UV) disinfection robot for corporate offices, warehouses, and other work spaces. The new robot is an evolution of a research project Ava Robotics conducted with the Massachusetts Institute of Technology and the Greater Boston...



9 robotics projects focusing on COVID-19

By Steve Crowe | September 24, 2021

Listen to this article

0:00 / 0:00

Voced by Amazon Polly



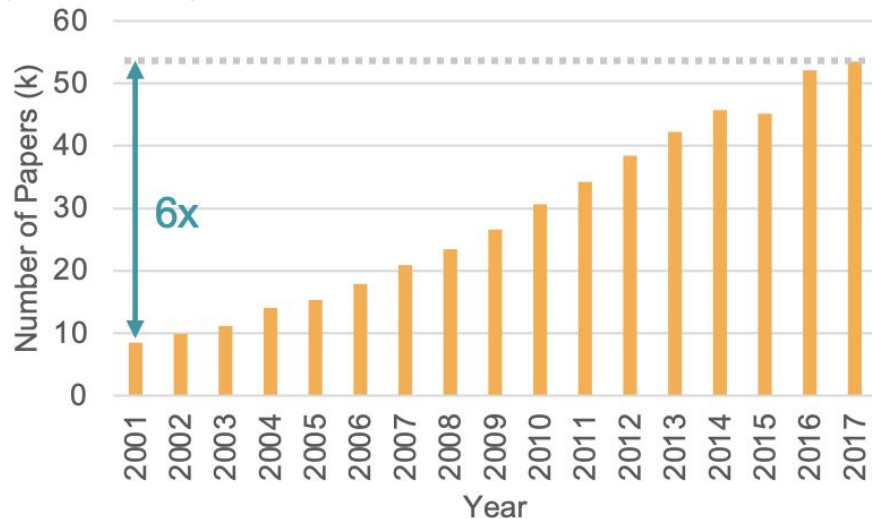
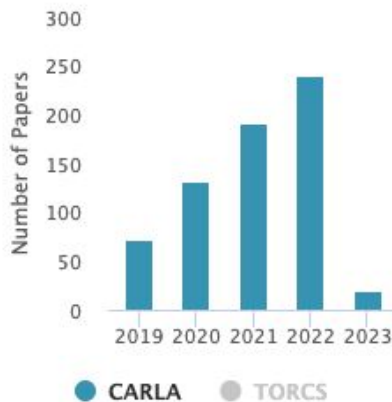
<https://www.therobotreport.com/tag/coronavirus/>

Now is a good time to study and work on Embodied AI

- The research community is rapidly growing
 - Publications at top conferences are on the rise!
 - ICRA, CoRL, RSS, IROS, etc.
 - Papers using 3D simulator is rapidly increasing

Why conferences are better than journals (most of the time)?
Because the research happens fast...

Usage 



Number of paper on robot learning

Embodied AI is challenging!

From DARPA Robotics Challenge 2015

A 3-stage competition to develop robot assistants. Tasks involved:

1. Robot drives through an obstacle course to destination and exit car
2. Enters a building through door
3. Turns a valve
4. Cuts a hole in a wall using a tool
5. Climb stairs to finish

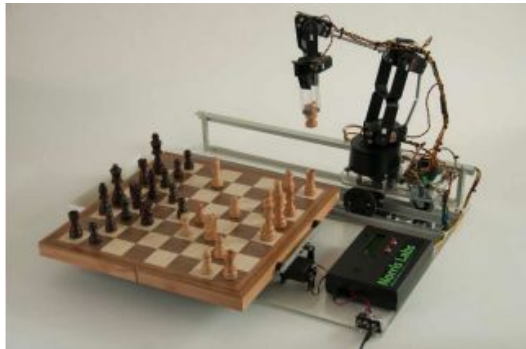


Only 3 teams successfully finished all the tasks

Embodied AI is challenging!

You may hear of “The Moravec's paradox” in the robotics literature

- The Moravec's paradox (1988):
 - “it is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility”



Still true in some sense nowadays. You can ask ChatGPT to write a convincing story but it could not do your laundry

Embodied AI is challenging!

You may hear of “The Moravec's paradox” in the robotics literature

- The Moravec's paradox (1988):
 - “it is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility”



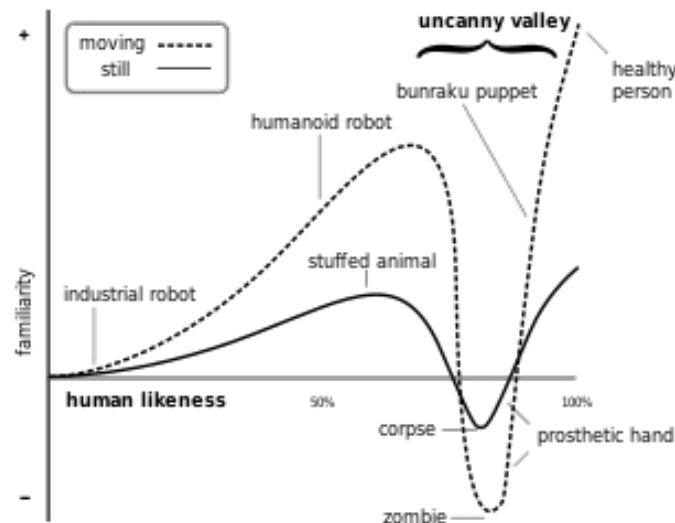
Boston Dynamics has the most advanced highly-agile robots. But they are not selling it (and at an affordable price) at any time soon.

Embodied AI is challenging!

You may also hear of The uncanny valley (恐怖谷理论)

- The uncanny valley (恐怖谷理论):

- "...a hypothesized relation between an object's degree of resemblance to a human being and the emotional response to the object"
- It means that if you want to build humanoid robot that looks like a human, it needs to be very good otherwise people will hate it
 - At some point we will need to consider the impact of human-computer interaction



As the robots become more like human, our familiarity with it may suddenly drops to a negative level

Agenda

- A bit about me...
- What is Embodied AI?
- History of Robotics
- Why study Embodied AI now?
- **Course content overview**
- Logistics

Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors



Robot Systems

Robot embodiment.
3D simulations.
Sim2real

Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors

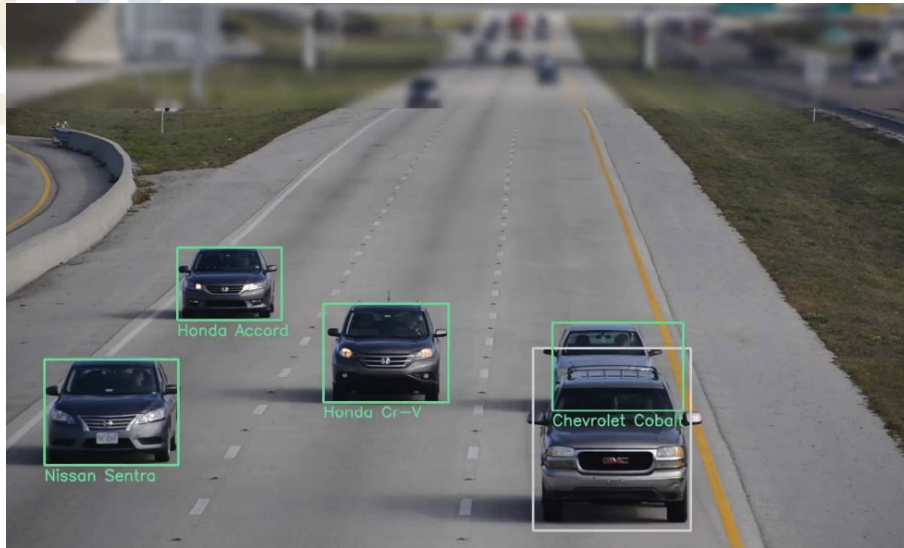


Robot Systems

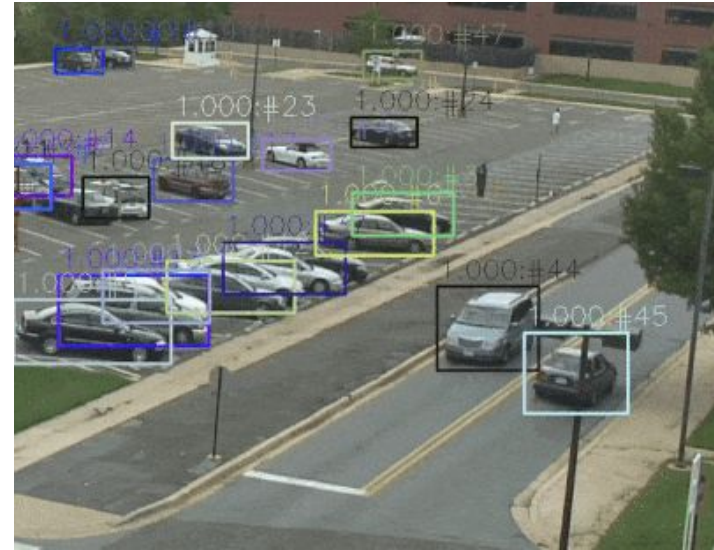
Robot embodiment.
3D simulations.
Sim2real

Embodied AI - Perception

- Typical tasks include Object detection & tracking



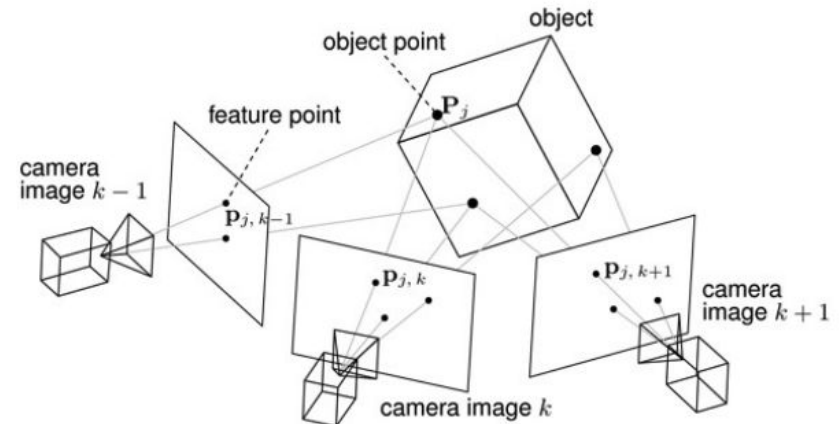
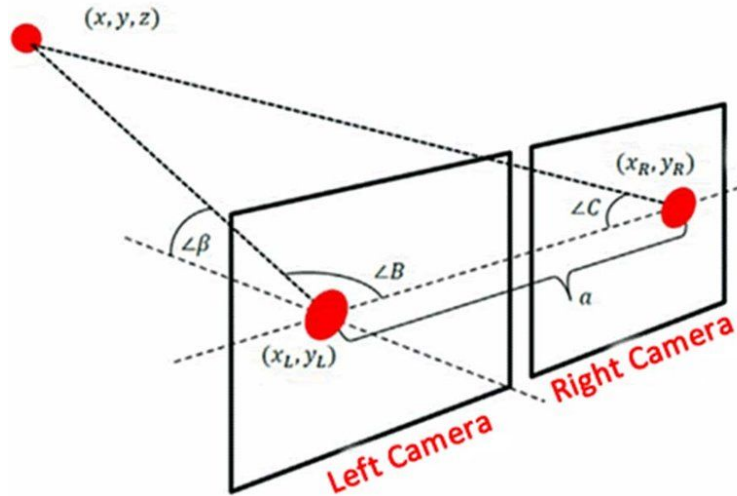
Drawing bounding boxes of different classes of objects



Tracking: Associate an ID for each object

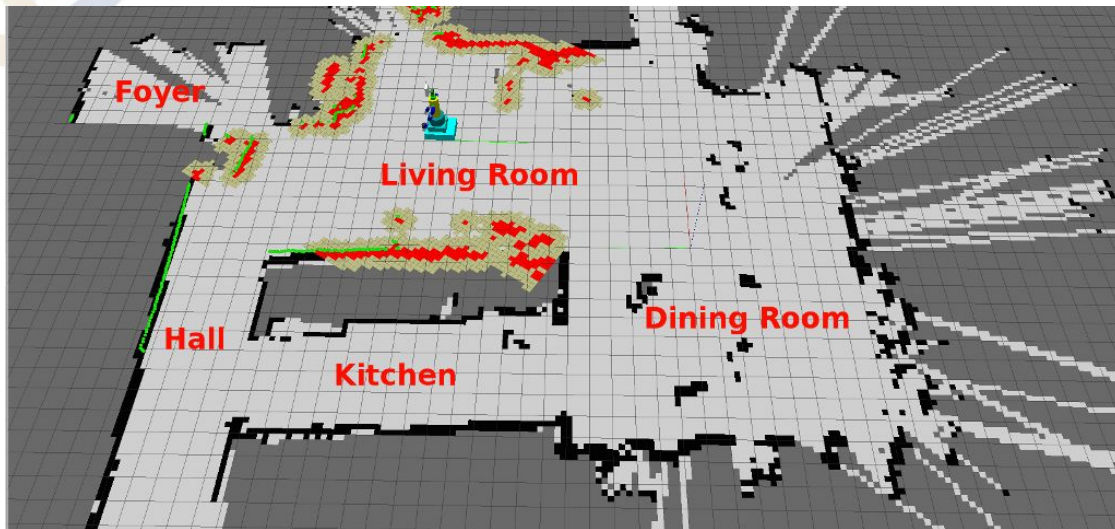
Embodied AI - Perception

- Stereo Vision
 - Can be used to estimate depth using geometrics of multiple cameras (will cover in this course)
 - And used for 3D reconstruction



Embodied AI - Perception

- Localization and mapping
 - Important for indoor robots
 - Could be done explicitly like in SLAM or implicitly



Goal: find a sofa.

history
modeling



Recursive
Implicit Map

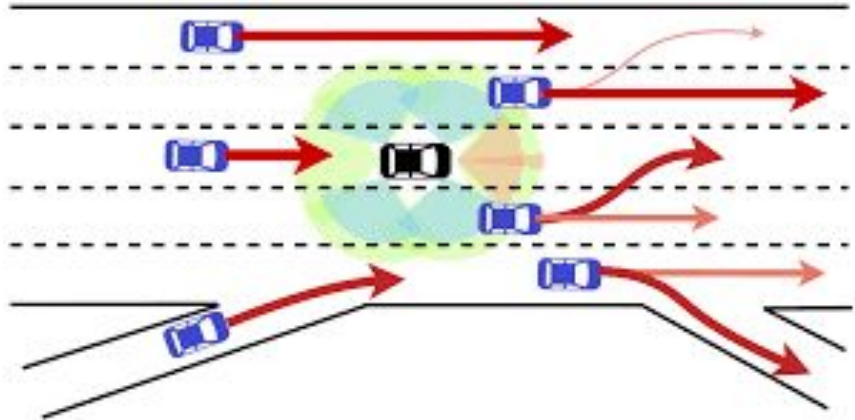
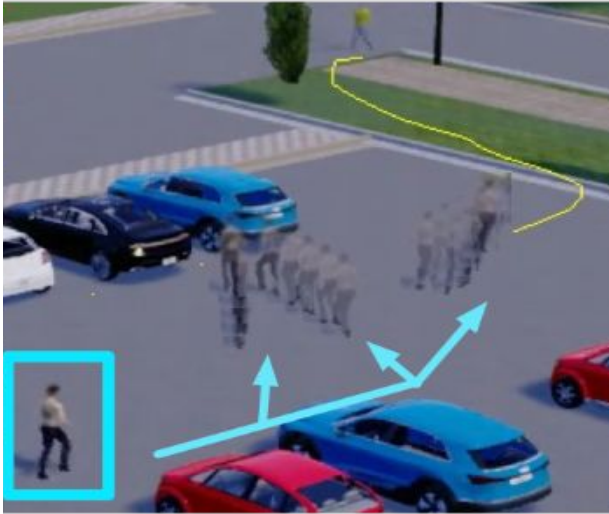
action
prediction



*Object Goal Navigation with
Recursive Implicit Maps. IROS 2023

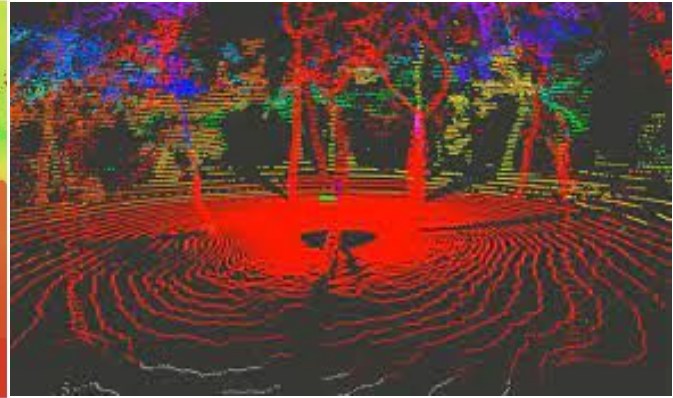
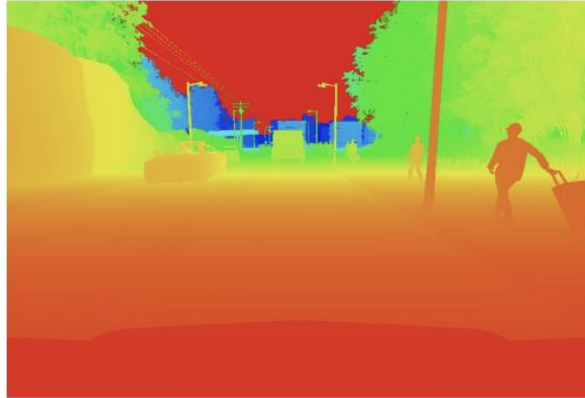
Embodied AI - Perception

- Trajectory Prediction
 - Predict the future trajectory/locations of vehicles and pedestrians
 - Usually within 5 seconds in the future



Embodied AI - Perception

- Multimodal Sensor Fusion
 - Combine information from RGB, Depth, and point clouds (LIDAR)



Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors

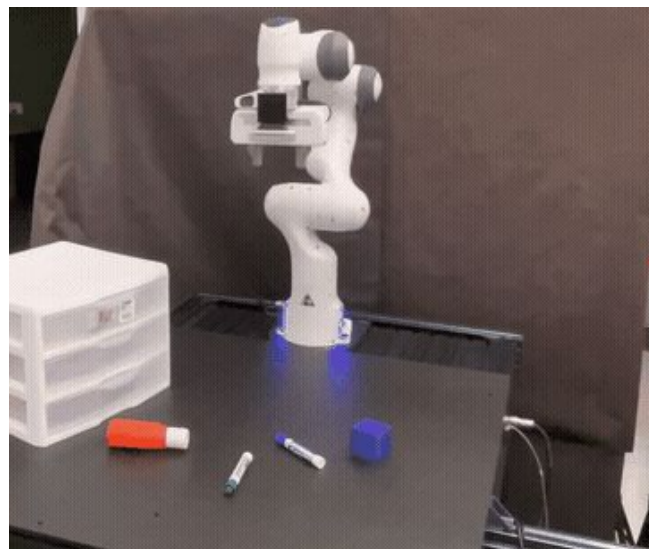


Robot Systems

Robot embodiment.
3D simulations.
Sim2real

Embodied AI - Decision Making

- Motion Planning and Control
 - Model outputs future motion plans for the robot to execute



*RVT: Robotic View Transformer for 3D Object Manipulation. CoRL 2023

Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors



Robot Systems

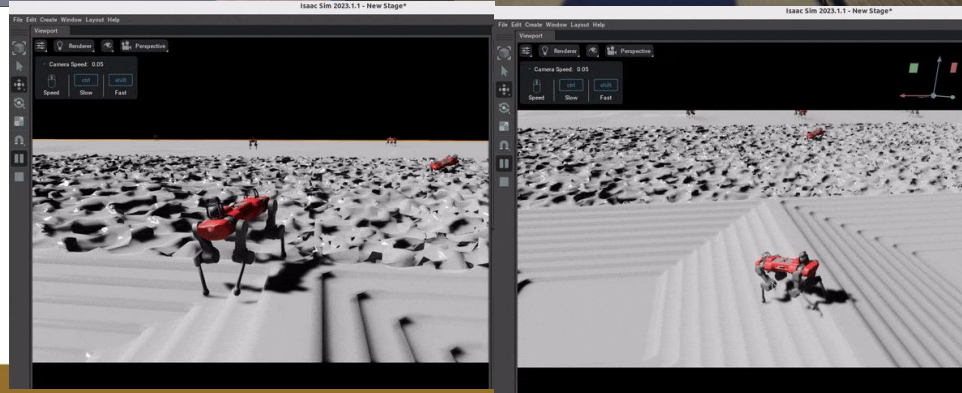
Robot embodiment.
3D simulations.
Sim2real

Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research



* CARLA simulator
(based on UE4):
<https://carla.org/>
<https://leaderboard.carla.org/>



* Nvidia Isaac Sim:
<https://developer.nvidia.com/isaac-sim>

Embodied AI - Robot Systems

3D simulator
can be very
realistic

This is from
CARLA -
Specialized
for
self-driving
task

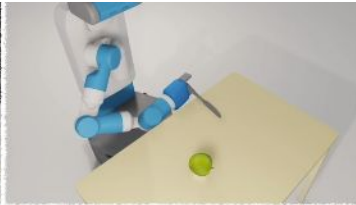


Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research



Wiping Table with Cloth



Slicing Apple with Knife

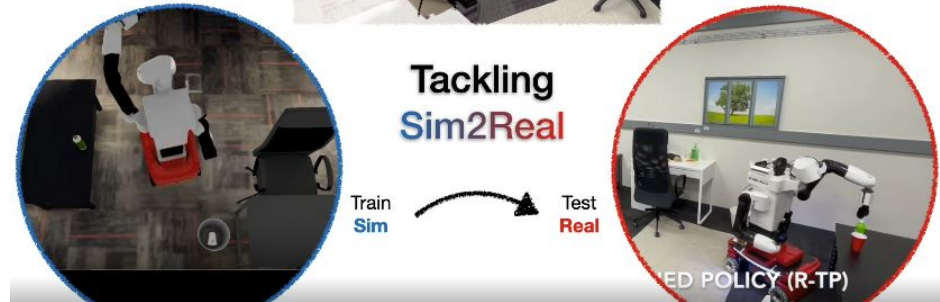
Complex Robot Behaviors

Ecological Environments

Setting Dinner Table



Cleaning Bathroom

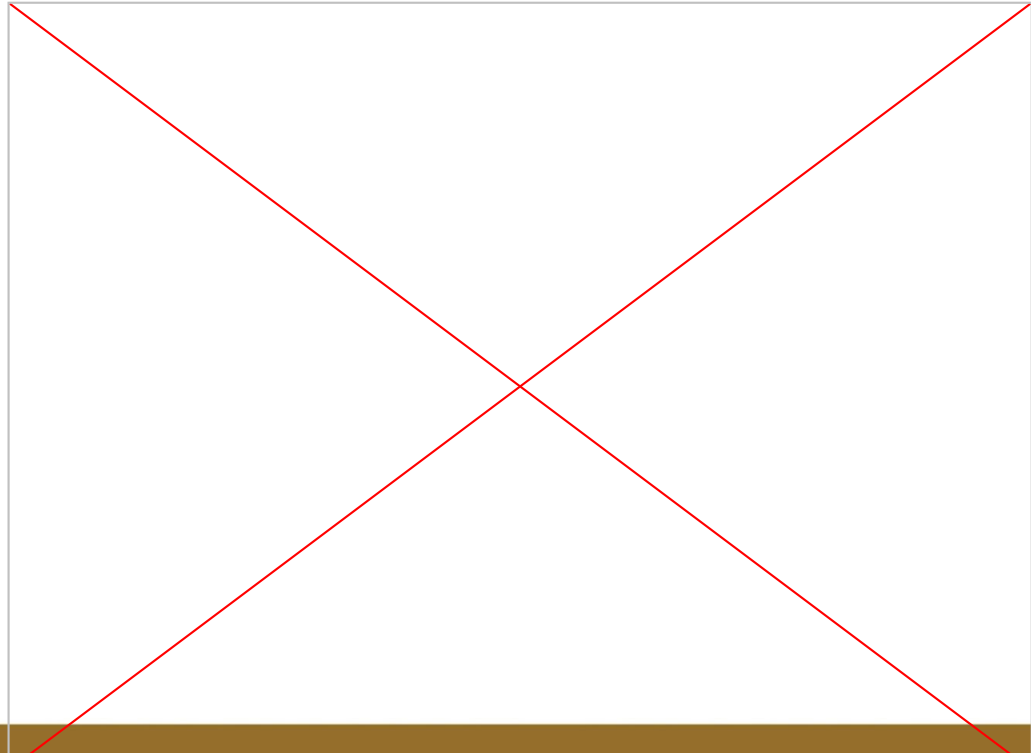


* BEHAVIOR-1K (Based on **Isaac Sim**): <https://behavior.stanford.edu/behavior-1k>

Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research

This is from BEHAVIOR-1k -
Specialized for everyday task
robots



Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors



Robot Systems

Robot embodiment.
3D simulations.
Sim2real

Course Plan

- Machine Perception
 - Robotic Sensors & Computer Vision
 - Visual Recognition
 - Object Tracking
 - Sensor Fusion - BEV
 - Localization and Mapping
- Robot Systems
 - 3D Simulators
 - PyTorch
 - Robot Operating System (ROS)
- Decision Making
 - **Embodied Navigation**
 - Reinforcement Learning, etc.
 - **Embodied Interaction and Manipulation**
 - VLA, etc.

Course Plan

- Course evaluation overview
 - 10 quizzes, multiple-choice questionnaires
 - Project 1: Paper Presentation
 - Project 2: Egocentric Object Detection
 - Project 3: Robot Navigation
- Requirements
 - Python programming
 - Basic calculus, linear algebra, probability theory
 - You should really be using Ubuntu (not Windows)
 - Most codes/tools are based on that
- Hardware requirement
 - For project 3, we need a machine with at least 8GB GPU
 - So at least 3060 TI / 4060 laptop (don't buy 50x0 series GPU)
 - If you don't have any GPU at all, I **suggest you drop the course**
 - Otherwise do not say it is unfair and give me a low score in the SFQ! :P
 - There some free resources, but I just don't think it is not enough/convenient to run Sim
 - Google Colab, Kaggle free GPU
 - 阿里云: <https://university.aliyun.com/> free ¥300 RMB credit

Course Objectives

- Describe key principles, concepts, models and techniques in Embodied AI
 - Understand technical challenges and recent research progress
- Implement algorithms introduced in class to solve small real-world problems
 - Familiarity with Embodied AI systems like ROS and 3D simulators
- Develop novel methods for Embodied AI applications
 - Navigation
 - Manipulation

Course Objectives

- Develop novel methods for Embodied AI applications
 - By the end of the course you can build your navigation robot dog that can avoid humans!

Falcon in diverse scenarios (Social-HM3D)

Third-person View

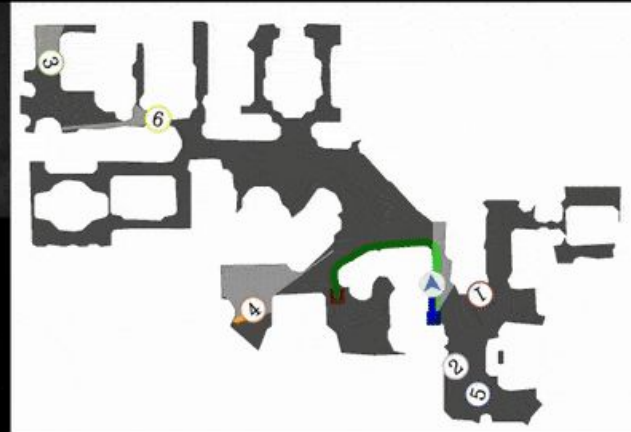


Depth



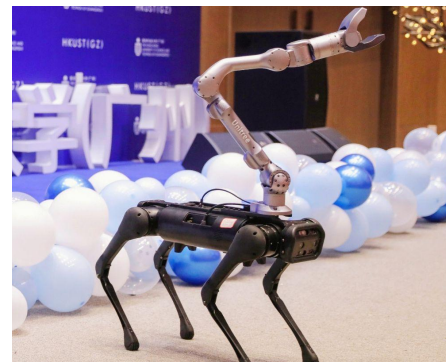
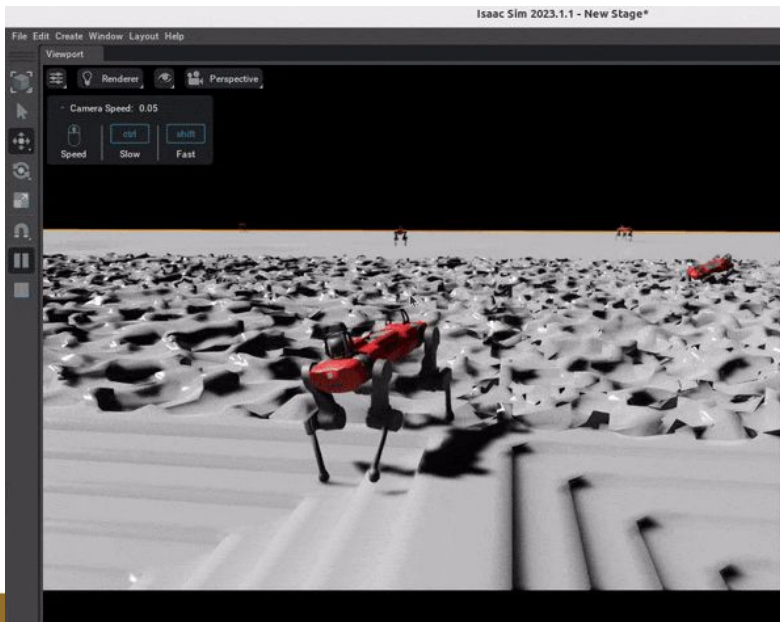
Falcon
(ours)

Top Down Map



Course Objectives

- Develop novel methods for Embodied AI applications
 - And if you are motivated and trained a robot dog with arm in Isaac Sim, you can use the robot from my lab to deploy your model!



Unitree Aliengo + Z1 arm:
You can search for research done
on this platform yourselves

Agenda

- A bit about me...
- What is Embodied AI?
- History of Robotics
- Why study Embodied AI now?
- Course content overview
- **Logistics**

Logistics: Part 1

- Course website
 - <https://hkust-aiaa4220.github.io/fall2025-L01/index.html>
 - You can see the **course schedule**, **homework links**, **download slides**, **policy**, etc.
 - The slides will be available before each class

Introduction to Embodied AI

AIAA 4220, Fall 2025

Instructors: [Junwei Liang](#)



Home Schedule	
Lecture:	Date and Time: 01:30PM - 04:20PM, Monday Location: Rm 149, E1 Websites: Canvas Zoom: 628-334-1826 PW: 12345 (In-person expected) (Recordings will be available after each class)
Course Description:	This course aims to provide students with key principles and algorithms to build modern Embodied AI systems. Key topics include machine perception, embodied navigation and manipulation algorithms. Through this course, students will learn and practice the foundational principles, techniques, and tools to build new Embodied AI systems.
Target Audience/ Prerequisites:	This is a course primarily for junior and senior students.
Course Work:	Grading In-Class Quizzes Project 1: Presentation Grading will be based on 10 in-class quizzes and attendance (30%), one presentation project (20%), one machine perception homework (20%) and one navigation homework (30%). • The in-class quizzes are designed to complement the lectures and making sure you understand the key knowledge about the course. • There are 10 in-class quizzes over the semester. The attendance is taken when you complete the quiz in-class. • All quizzes will be 9 multiple-choice questions. You will have 10 minutes (tentative) to finish. • You will select 1-2 papers from our list of Embodied AI research to present. This should show that you understand the paper, and you could present it to students who do not know the paper well. • Can be done in groups of 1-3 people. • Project 1 is worth 20% credit. You will be graded based on Understanding & Content (40%), Presentation & Communication (30%), Critical Analysis & Discussion (30%), by the teacher, TAs, and peers.

Logistics: Part 1

- Course website
 - <https://hkust-aiaa4220.github.io/fall2025-L01/index.html>
 - You can see the **course schedule**, **homework links**, **download slides**, **policy**, etc.
 - The slides will be available before each class
- Canvas
 - <https://hkust-gz.instructure.com/courses/596>
 - Course announcement will be posted here
 - Use the discussion section for questions and discussions
 - **We will use Canvas for quizzes**
- Hardware requirement (Again!)
 - For project 3, we need a machine with at least 8GB GPU
 - So at least 3060 TI / 4060 laptop (don't buy 50x0 series GPU)
 - If you don't have any GPU at all, I **suggest you drop the course**
 - Otherwise do not say it is unfair and give me a low score in the SFQ! :P
 - There some free resources, but I just don't think it is not enough/convenient to run Sim
 - Google Colab, Kaggle free GPU
 - 阿里云: <https://university.aliyun.com/> free ¥300 RMB credit

Logistics - Computation Resources from Aliyun

- Aliyun has provided our university with free or discounted computation resources
 - <https://university.aliyun.com/>
 - 300 RMB voucher
 - 70% off for orders up to 5000 RMB/year
 - You can directly apply with your hkust-gz email

云工开物
阿里云高校计划，助力高校科研与教育加速，让每位中国在校大学生真实受益于普惠算力

助力高校师生云上“创世界”

“云工开物”取自中国古代科技史上里程碑式的科学著作《天工开物》，反映了伏羲、和与空想、要实干的魄力。云计算作为推动科研新范式的关键引擎，将助力现实。“云工开物”将倾力支持高校教师云上科研提速，取得有世界级影响力的成果在云上探索更多可能性，创造出更精彩的未来世界。

面向高校学生 | **面向高校教师**

高校学生权益
面向中国高校学生，立即申领权益开启云上实践 [活动规则](#)

高校学生通用权益
完成高校学生认证即可领取，其中港澳台地区近期开启

立即领取

¥300
无门槛优惠券

适用范围：阿里云全量公共云产品（特殊商品除外）
使用说明：自领取之日起，1年内有效，如订单退款，优惠券不

合作高校学生专属权益
[合作高校](#) 学生完成学生认证即可领取，可与300元优惠券叠加使用

立即领取

3折
折扣优惠

适用范围：阿里云全量公共云产品（特殊商品除外），封顶5000元/年
使用说明：自领取之日起，1年内有效，如订单退款，仍占用优

Logistics: Part 2

- Course aims to give you more hands-on experience
 - Means coding and experimenting
- Language of choice: Python
- Toolkit of choice: PyTorch / Habitat Sim
 - You are welcome to use other languages/toolkits, but the TAs will not be able to help
- Participation
 - Interrupting (by raising a hand) to ask your question is strongly encouraged
 - Asking questions later on Canvas (or in class) is also great
 - Interaction improves learning (both in-class and at my office hours)
- Office hours
 - Rm 304, E4
 - Monday 4:30 - 5:30 pm.

Grading

- This course utilizes an **absolute grading** system as follows
 - A:[100,85] B:(85,70] C (70,55] D:(55,40] F:(40,0]
 - Subgrades (e.g, A-, B+, etc,) will be assigned for every 5 points within these ranges.
 - If the grades are too low, we may **curve up** the whole class
- In-class Quizzes (25%)
- Project 1: Paper Presentation (20%)
- Project 2: Machine Perception Homework (20%)
- Project 3: Navigation Homework (30%)
- Attendance (5%)

Grading - Details

- In-class Quizzes (25%) + Attendance (5%) = 30%
 - Designed to complement the lectures and making sure you understand the key knowledge about the course
 - 10 individual assignment (so you need to complete this yourselves)
 - **Usually scheduled at the end of the first half of the class (around 2:30 pm.), due in-class**
 - All quizzes will be 9 multiple-choice questions. You will have 10 minutes (tentative) to finish.
 - **Will be posted on Canvas Quizzes section**
 - First quiz will be out on 9/8 (next Monday)

Grading - Details

- Project 1: Paper Presentation (20%)
 - You will select 1 paper from our list of Embodied AI research to present. This should show that you understand the paper, and you could present it to students who do not know the paper well. **You have 10 minutes to present and 5 minutes to handle at least one question (from TA/peers/teacher)**
 - Will be released on 9/8
 - Full schedule
 - P1 group due on 9/15
 - P1 presentation order release on 9/22
 - P1 presentation on 10/13
 - Can be done in groups of 1-3 people. **Score multiplier: 1.1x for solo, 1.05x for two people and 1.0x for three people group.**
 - Project 1 is worth 20% credit.
 - You will be graded based on Understanding & Content (40%), Presentation & Communication (30%), Critical Analysis & Discussion (30%), by the teacher, TAs, and peers.

Grading - Details

- Project 2: Machine Perception Homework (20%)
 - You will do an object detection homework.
 - Can be done in groups of 1-3 people.
 - You will submit to a public leaderboard. You will be evaluated based on model performance
 - Details to come. **Score cutoff, multiplier etc.**
 - Tentative schedule:
 - P2 & P3 release together on 9/29
 - Form a group for P2 & P3 by 10/13
 - Due 11/24. So you almost have two months' time to complete
 - Top-3 team will be invited to present their methods on 12/1. They get 4%/2%/1% extra credit.

Grading - Details

- Project 3: Navigation Homework (30%)
 - You will do a social navigation homework.
 - Can be done in groups of 1-3 people.
 - You will submit to a public leaderboard. You will be evaluated based on model performance
 - Details to come. **Score cutoff, multiplier etc.**
 - Tentative schedule:
 - P2 & P3 release together on 9/29
 - Form a group for P2 & P3 by 10/13
 - Due 11/24. So you almost have two months' time to complete
 - Top-3 team will be invited to present their methods on 12/1. They get 4%/2%/1% extra credit.

Grading - Details

- Late Policy
 - Homework assignments and course project results are worth full credit on the due date. Unless granted an extension in advance, it is worth at most 75% credit for the next 48 hours, at most 50% credit after that. If you need an extension, please ask for it as soon as the need for it is known. Extensions that are requested promptly can be granted more liberally. You must turn in all assignments.

Grading - Details

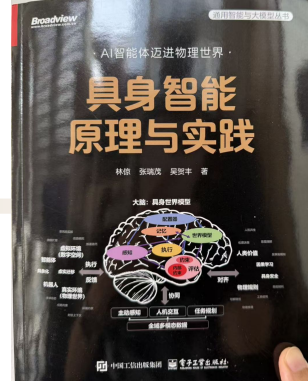
- Collaboration
 - We encourage collaboration between students and studying materials in groups when the purpose of this is to facilitate learning, not to circumvent problems. It is allowed to seek help from other students in understanding the material needed to solve a particular problem. However, students must submit individual material and solutions, unless otherwise specified. Students should declare any collaboration on the first page of homework assignments (or equivalently on exercises). If the instructors believe the collaboration is improper, your grade may be affected

Grading - Details

- Use of LLM/VLMs (ChatGPT, Gemini, etc.)
 - We encourage you to use ChatGPT to solve the homework assignment problems (because likely in the future everyone will use it for coding). However, you will need to declare using ChatGPT/LLMs in the homework README, as well as the prompts you use
 - **You are not allowed to use LLM for in-class test!!!**

More Course Resources

- Textbooks
 - Machine Learning: A Probabilistic Perspective - Kevin P. Murphy
 - Multiple View Geometry in Computer Vision - Richard Hartley and Andrew Zisserman
 - Introduction to Autonomous Mobile Robots - Illah Reza Nourbakhsh and Roland Siegwart
 - Chinese textbook: 具身智能原理与实践 by my professor friends
- Workshops and Competitions
 - Precognition @CVPR [\[link\]](#)
 - Workshop on Autonomous Driving @CVPR [\[link\]](#)
 - CARLA Autonomous Driving Challenge @NeurIPS [\[link\]](#)
- Reading Materials
 - Nice talks
 - [A Path Towards Embodied AI - Yann LeCun](#) [\[link\]](#)
 - Robotics Professor Answers Robot Questions From Twitter - Henny Admoni [\[link\]](#)
 - Other Open Courses
 - Stanford AA 274a
 - CMU 16-311, 11-851
 - Washington CSE 478
 - Texas CS 391R
- **More online tutorial videos/Important knowledge about Embodied AI by Prof. Junwei Liang**
 - 【Tencent Docs】Embodied AI学习指引 for UG
 - <https://docs.qq.com/sheet/DREN4UUpyaVBvbGdL?tab=qgffmc>



Your ToDOs

- Check course webpage, Canvas
 - <https://hkust-aiaa4220.github.io/fall2025-L01/index.html>
- Review linear algebra, calculus and probability
- Review python coding, numpy, Ubuntu system
- Checkout 3D simulators and play around
 - <https://carla.org/>
 - <https://behavior.stanford.edu/>
 - <https://developer.nvidia.com/isaac-sim>
- Start finding group members to pair with others in course projects!



Thank you for deciding to take this journey of Embodied AI with us

Hope you can learn something and get a good grade!

- Be aware: most likely that it would be challenging to get full grade
- By if you all get above 90 then all of you get A

Any questions?

On what laptop/desktop GPU to buy?